THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

May 23, 2017 2:30 p.m. – 5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL FIVE questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (28 points)

In this question, you may use without proof the following if necessary:

- $\sqrt{2}$ is irrational.
- The sum/difference/product/quotient of two rational numbers is rational.
- The composition of two injective/surjective/bijective functions is injective/surjective/bijective respectively.

Prove or disprove each of the following statements:

(a) For any $a, b \in \mathbb{Q}$ with $a < b$, there exists $c \in (a, b)$ such that $c \notin \mathbb{Q}$.

(b) For any denumerable set A, there exists a function from A to $\mathbb{N}$ that is injective but not surjective.

(c) There exist nonempty sets A, B and C such that $|A| < |B|$, $|B| \leq |C|$ and $|C| \leq |A|$.

(d) $|[-4, 4]| = |(-8, -1) \cup (10, 20)|$.

(e) Let A be a nonempty set, and let P_1 and P_2 be two partitions of A. Define

$$P = \{S \in \mathcal{P}(A) \setminus \{\emptyset\} : S = S_1 \cap S_2 \text{ for some } S_1 \in P_1 \text{ and } S_2 \in P_2\}.$$

Then P is a partition of A.

2. (16 points)

Define the n-th *Fermat number* F_n by

$$F_n = 2^{(2^n)} + 1 \quad \text{for any integer } n \geq 0.$$

(a) Prove by induction that

$$F_0 F_1 \cdots F_{n-1} = F_n - 2 \quad \text{for any integer } n \geq 1.$$

(b) Prove that no two Fermat numbers have a common factor greater than 1.

(c) Use (b) to show that there are infinitely many prime numbers.

3. (16 points)

Let A be a nonempty set and R be a relation on A. We say that R is *anti-symmetric* if for any $x, y \in A$,

$$xRy \text{ and } yRx \implies x = y.$$

We say that R is an *ordering* if R is reflexive, anti-symmetric and transitive.

(a) Let S be a relation on $\mathbb{R}^2$ defined by

$$(x_1, y_1)\, S\, (x_2, y_2) \quad \text{if} \quad \sqrt{x_1^2 + y_1^2} \leq \sqrt{x_2^2 + y_2^2}.$$

<u>Prove or disprove</u> that S is an ordering.

(b) Let

$$B = \{(x, y) \in \mathbb{R}^2 : (x, y) = (0, 0) \text{ or } x > 0\}.$$

Let T be a relation on $\mathbb{R}^2$ defined by

$$(x_1, y_1)\, T\, (x_2, y_2) \quad \text{if} \quad (x_1 - x_2, y_1 - y_2) \in B.$$

<u>Prove or disprove</u> that T is an ordering.

(c) Prove that for any nonempty set A, there is exactly one relation which is an ordering as well as an equivalence relation.

4. (20 points)

(a) Prove that for any $a \in \mathbb{Z}$, if $\dfrac{a}{3^k} \in \mathbb{Z}$ for any $k \in \mathbb{N}$, then $a = 0$.

(b) Prove that for any $a \in \mathbb{Z}$, $a^3 \equiv 0 \pmod 3$ if and only if $a \equiv 0 \pmod 3$.

(c) Suppose that there exist $a, b, c \in \mathbb{Z}$ that satisfy the equation

$$a^3 + 3b^3 = 9c^3.$$

Use (a) and (b) to prove that the only integer solution is $a = b = c = 0$.

(d) Prove that the only rational solution to the equation in (c) is also $a = b = c = 0$.

5. (20 points)

Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a function.

(a) Prove that if f is injective, then $X = f^{-1}(f(X))$ for any subset $X \subseteq A$.

(b) Prove that if f is surjective, then $f(f^{-1}(Y)) = Y$ for any subset $Y \subseteq B$.

(c) Define a function $F : \mathcal{P}(A) \longrightarrow \mathcal{P}(B)$ by $F(X) = f(X)$ for any $X \in \mathcal{P}(A)$.

(i) Prove that f is injective if and only if F is injective.

(ii) Prove that f is surjective if and only if F is surjective.

(d) Prove that if A is numerically equivalent to B, then $\mathcal{P}(A)$ is numerically equivalent to $\mathcal{P}(B)$.

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